YEAR 10 SCIENCE

EARTH AND SPACE SCIENCE
(General)

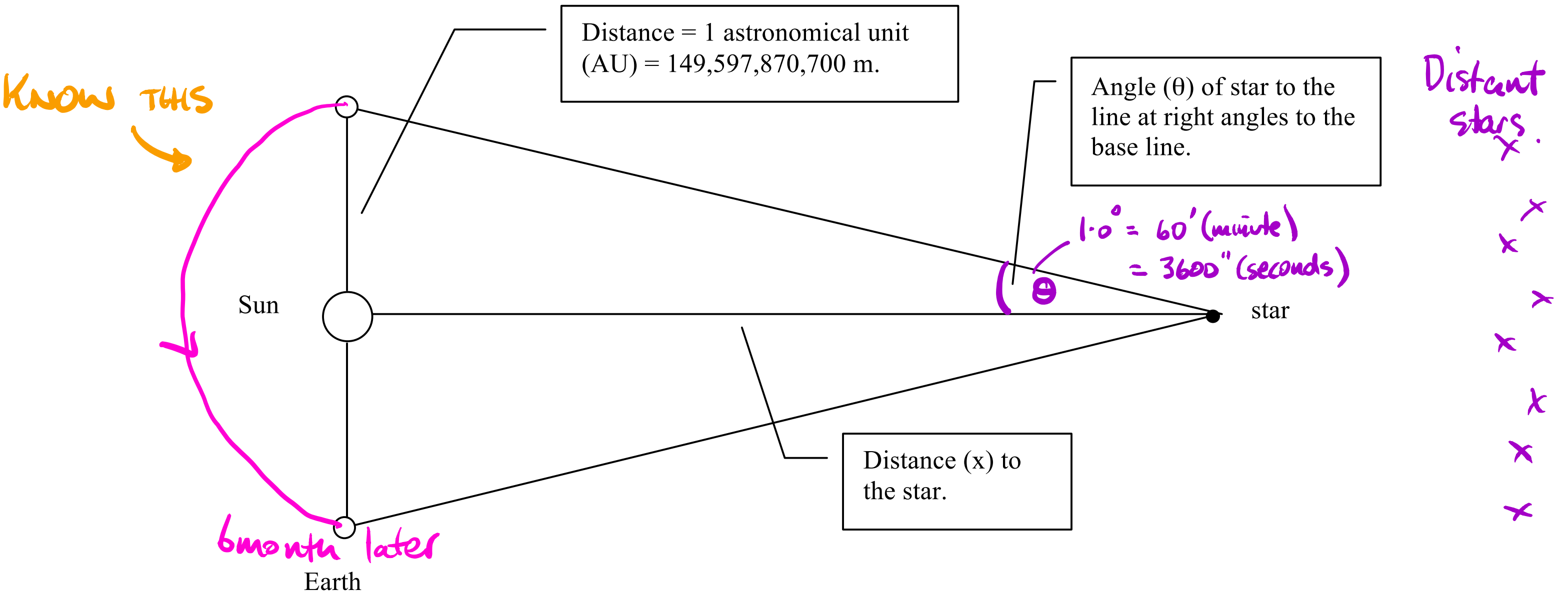
ASTRONOMY

NAME: ______________________________

MEASURING DISTANCES TO STARS

Close Stars

- ***Parallax method*** is used.
- Stars are observed 6 months apart.
- Close stars appear to move but distant stars do not.
- Trigonometry is used to calculate the distance.



To calculate the distance x to a star :

$$\boxed{\tan\theta = \frac{149{,}597{,}870{,}700\ \text{m}}{x}}$$

Assume the parallax angle measured to a star is 0.0136 °.

$$\Rightarrow \qquad \tan 0.0136° = \frac{149{,}597{,}870{,}700}{x}$$

$$\Rightarrow \qquad \underline{x = 6.032 \times 10^{14}\ \text{m}}$$

- The ***parsec*** (pc) is used as a measure of intergalactic distances.
- Using the above calculation, with parallax angle q = 1.0 second of arc (1/3600 °):

1.00 parsec = 3.085678×10^{16} m = 3.261564 light years

Distant Stars

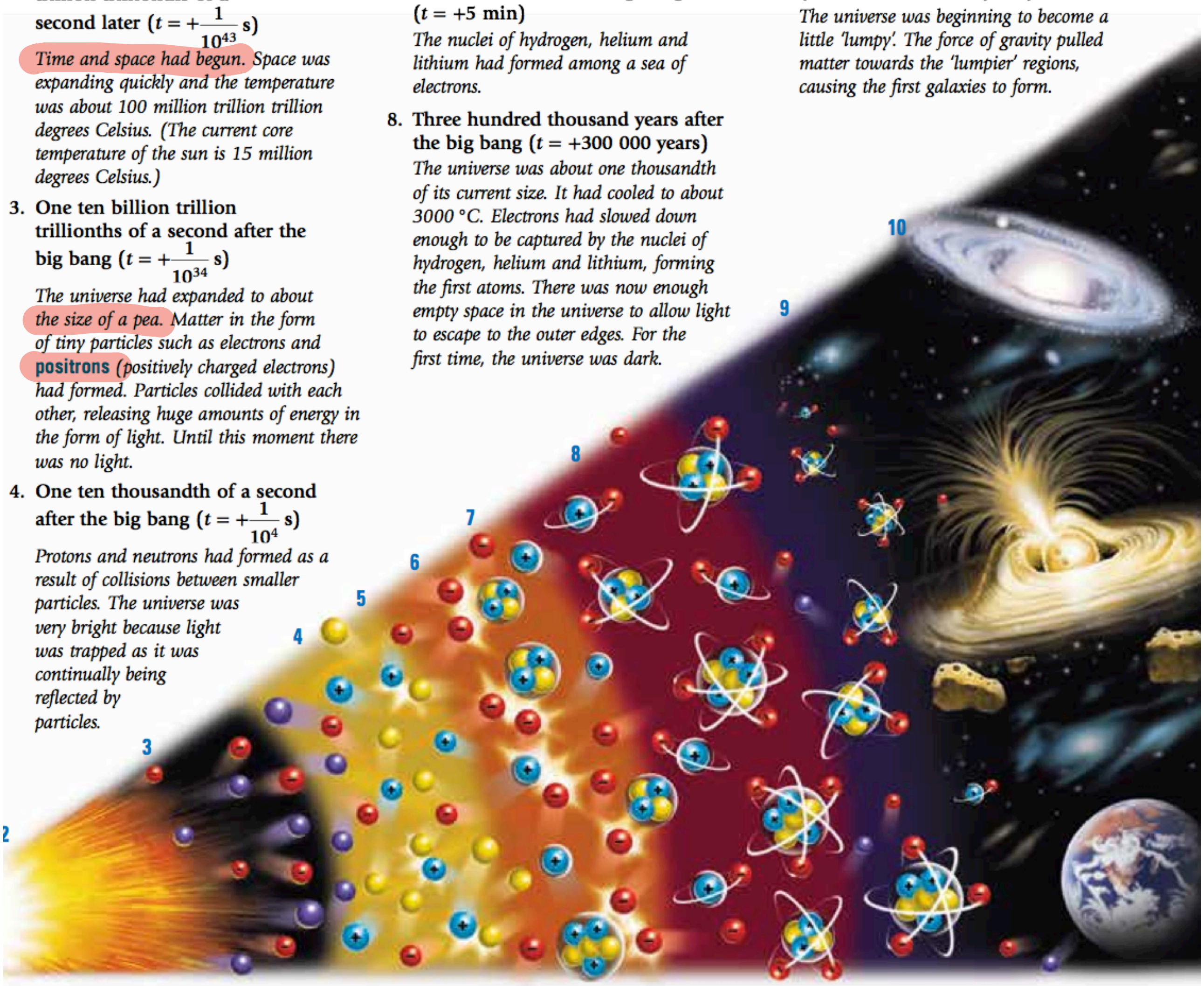
- Distant stars do not appear to move very much when viewed by parallax method.
- The ***apparent brightness*** of the stars is used to measure their distances.
- ***Cepheid variables*** vary in brightness over a set period of time and are used to estimate distances to galaxies.

ORIGIN OF THE UNIVERSE

When and how did the universe begin? Was there a beginning? Perhaps it was always there. If there was a beginning, will there be an end? The study of the answers to these questions is called ***cosmology***.

Following Edwin Hubble's discoveries about the expanding universe, two major theories about the beginning of the universe became popular — the **big bang theory** and the **steady state theory**.

1. **The big bang ($t = 0$)**
   It's hard to imagine, but at this moment there was no space and no time. All that existed was energy, which was concentrated into a single point called **singularity**.
2. **One ten million trillion trillion trillionths of a second later ($t = +\frac{1}{10^{43}}$ s)**
   Time and space had begun. Space was expanding quickly and the temperature was about 100 million trillion trillion degrees Celsius. (The current core temperature of the sun is 15 million degrees Celsius.)
3. **One ten billion trillion trillionths of a second after the big bang ($t = +\frac{1}{10^{34}}$ s)**
   The universe had expanded to about the size of a pea. Matter in the form of tiny particles such as electrons and **positrons** *(positively charged electrons) had formed. Particles collided with each other, releasing huge amounts of energy in the form of light. Until this moment there was no light.*
4. **One ten thousandth of a second after the big bang ($t = +\frac{1}{10^{4}}$ s)**
   Protons and neutrons had formed as a result of collisions between smaller particles. The universe was very bright because light was trapped as it was continually being reflected by particles.
5. **One hundredth of a second after the big bang ($t = +\frac{1}{100}$ s)**
   The universe was still expanding and cooling rapidly. It had grown to the same size as our solar system, but there was still no such thing as an atom.
6. **One second after the big bang ($t = +1$ s)**
   The universe was probably more than a trillion trillion kilometres across. It had cooled to about ten billion degrees Celsius.
7. **Five minutes after the big bang ($t = +5$ min)**
   The nuclei of hydrogen, helium and lithium had formed among a sea of electrons.
8. **Three hundred thousand years after the big bang ($t = +300\ 000$ years)**
   The universe was about one thousandth of its current size. It had cooled to about 3000 °C. Electrons had slowed down enough to be captured by the nuclei of hydrogen, helium and lithium, forming the first atoms. There was now enough empty space in the universe to allow light to escape to the outer edges. For the first time, the universe was dark.
9. **Two hundred million years after the big bang ($t = +200\ 000\ 000$ years)**
   The first stars had appeared as gravity pulled atoms of hydrogen, helium and lithium together. **Nuclear reactions** *took place inside the stars, causing the nuclei of the atoms to fuse together to form heavier nuclei. Around some of the newly forming stars, swirling clouds of matter cooled and formed clumps. This is how planets began to form.*
10. **One billion years after the big bang ($t = +1\ 000\ 000\ 000$ years)**
    The universe was beginning to become a little 'lumpy'. The force of gravity pulled matter towards the 'lumpier' regions, causing the first galaxies to form.



13.7 billion yrs?

THE BIG BANG

According to the most commonly accepted theory among cosmologists, the universe began about 15 billion years ago with a ‘big bang’. This is better described as "The Big Expansion".

At the beginning, all that existed was energy, which was concentrated into a single point called a ***singularity***.

The Einstein connection

"General relativity + gravity" – mention it.

The singularity before the Big Bang was not ‘nothing’. It was a huge amount of energy (with no mass) concentrated into a tiny, tiny point.

Einstein proposed that energy could be changed into matter. His equation $E = mc^2$ describes the change.

in the first 5 minutes.

> E represents the amount of energy in joules.
> m represents the mass in kilograms.
> c is the speed of light in metres per second (300 000 000 m/s).

Einstein’s equation also describes how matter can be changed into energy. That is what happens in nuclear power stations, nuclear weapons and during the fusion process in stars.

HOW ABOUT THAT!

The big bang theory was first proposed in 1927 by Georges Lemaitre, a Catholic priest from Belgium. But it wasn’t called the ‘big bang theory’ then. Ironically, the name ‘big bang’ was invented by Fred Hoyle, one of the developers of the steady state theory. He used the name to try to ridicule the cosmologists who proposed the big bang theory.

In 1929, Edwin Hubble’s observations and law provided convincing evidence to support Lemaitre’s proposal. In 1933, Lemaitre presented the details of his theory to an audience of scientists in California. Albert Einstein, by then recognised as one of the greatest scientists of all time, was in the audience. At the end of Lemaitre’s presentation Einstein stood, applauded and announced, ‘That was the most beautiful and satisfactory explanation of creation that I have ever heard’.

The Steady State Theory

Must mention it + explain it.

According to the Steady State Theory, which was proposed in 1948 by Fred Hoyle, there was no beginning of the universe. It was always there.

unchanging - looks the same everywhere

The galaxies are continually moving away from each other. In the extra space left between the galaxies, new stars and galaxies are created.

These new stars and galaxies replace those that move away, so that the universe always looks the same.

A huge debate between those who supported the Steady State Theory and those who supported the Big Bang Theory raged from 1948 until 1965. During that period, the evidence supporting the Big Bang Theory grew.

Supporting Evidence

- **The red shift**

The red shift provides evidence for an expanding universe. This evidence supports the Big Bang Theory and causes problems for those supporting the Steady State Theory.

A Steady State universe could expand only if new stars and galaxies replaced those that moved away. These young stars and galaxies have not been found by astronomers.

- **The elements**

The amounts of hydrogen and helium in the universe support the Big Bang Theory. The amount of helium in the universe is greater than that proposed by the Steady State theory, but it can be explained by its creation as a result of the Big Bang.

- **The afterglow**

When George Gamow and Ralph Alpher proposed their version of the Big Bang Theory in 1948, they calculated that the universe would now, about 15 billion years after creation, have a temperature of 2.7°C above ***absolute zero*** (about -270°C).

2·7° Kelvin

Gamow predicted that, because of its temperature, the universe would be emitting an 'afterglow' of radiation. This afterglow became known as cosmic microwave background radiation, which was discovered in 1965.

Its discovery put an end to the Steady State Theory, leaving the Big Bang Theory as the only theory supported by evidence currently available.

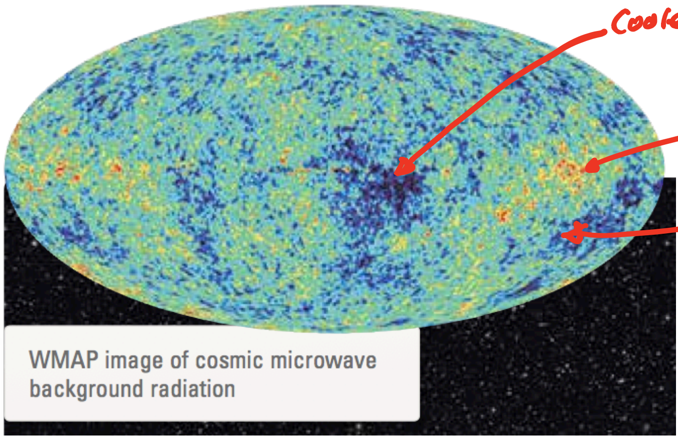
Mention the background radiation.

$-273°C = 0°K$

- **Cosmic Microwave Background Radiation**

The background radiation detected was released only 380 000 years after the Big Bang — the first radiation to escape.

The image shows how the temperature varied across the universe as it was 380 000 years after the Big Bang.



WMAP image of cosmic microwave background radiation

Cooler

warmer

Doesn't look the same everywhere

The blue parts of the map are the cooler regions. These regions were cool enough for atoms, and eventually galaxies, to form.

The red parts are warmer regions.

The map shows that galaxies are not evenly spread throughout the universe. They support the theory of an expanding universe that began with a Big Bang.

What is the future?

Will the expansion of the universe continue forever? If the universe does stop expanding, what will happen to it?

There are several competing theories about the answers to these questions.

One theory suggests that there is not enough mass in the universe for gravity to be able to pull it all back, so it will continue to expand forever.

Other theories suggest that the universe will eventually end. According to these theories, the end will come when:

- The universe snaps back onto itself in a ‘big crunch’ (the ***Big Crunch Theory***). The end result will be a single point — singularity.
  Some cosmologists believe that the Big Crunch will be followed by another Big Bang.

- The expansion of the universe continues and stars use up their fuel and burn out, causing planets to freeze (the ***Big Chill Theory***).
  The universe would then consist of scattered particles that never meet again.

- The universe rips itself apart violently as a result of expanding at an increasing speed (the ***Big Rip Theory***).
  According to this theory, the end of the universe will also be the end of time itself.

Worksheet 6.5

The big bang

Student: .... Class: ....

Flow charting the big bang

1. Complete the big bang flow chart by selecting the event from the list below that matches each step in the sequence and rewriting it in the correct box. Enter a title for each box that summarises each event.

$t = 0$	$t = 10^{-43}$ s
$t = 10^{-4}$ s	$t = 10^{-34}$ s
$t = 10^{-2}$ s	$t = 1$ s
$t = 3 \times 10^{5}$ yr	$t = 5$ min
$t = 2 \times 10^{8}$ yr	$t = 1 \times 10^{9}$ yr

Big bang events in random order:

A. Universe has cooled to 3000 °C; first atoms form.

B. Galaxies begin to form.

C. Time and space form; space expands rapidly.

D. Heavier matter forms — protons, neutrons.

E. Singularity exists — the big bang occurs.

F. Universe cools; reaches size of the solar system.

G. Atomic nuclei form.

H. First stars appear.

I. Light matter forms — electrons, positrons.

J. Universe has cooled to 10^{10} °C.

Evidence for the big bang

2. The following microwave map of the early universe (t = 380 000 years after big bang) was produced by a probe that orbited the Earth in 2001.

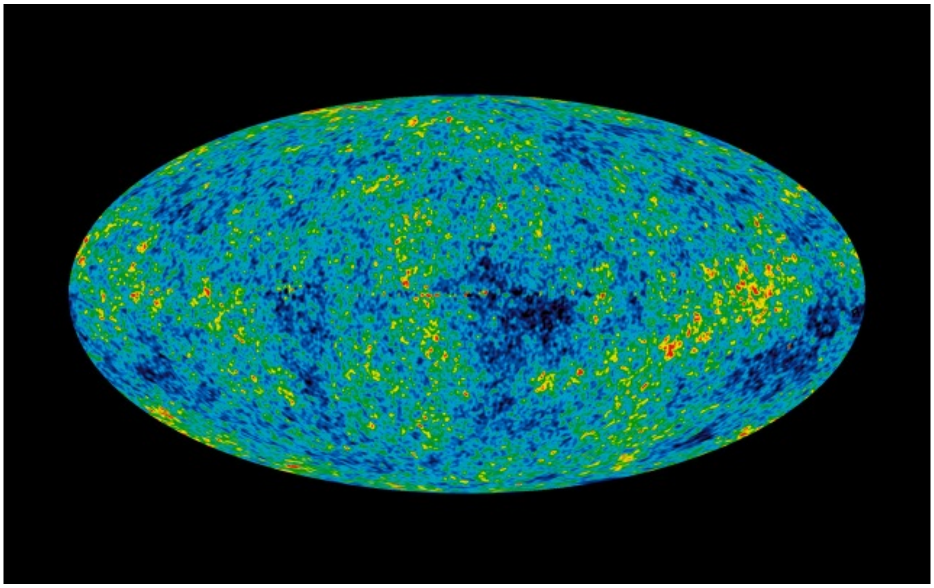


Photo: NASA

(a) The map shows that in the early universe matter was not uniformly distributed. Explain how this map supports the big bang theory.

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(b) In 1948, Gamow and Alpher predicted that microwave radiation should now fill the universe. They proposed that this radiation was the cooled 'afterglow' of the big bang. When was this microwave radiation first detected?

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DOPPLER EFFECT

- This affects both sound and light.
- On Earth, an approaching ambulance or police car with its siren sounding appears to change its frequency due to its movement.
- **Approaching** - the frequency sounds higher.

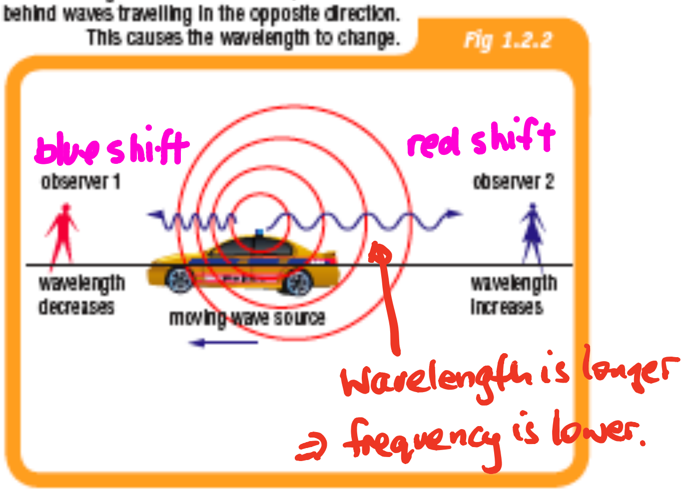
  => ***Wavelength is apparently smaller*** (since the speed of sound in the air remains unchanged.

Remember

- **Going away** - the frequency sounds lower.

  => ***Wavelength is apparently larger***.

A moving source of waves tends to catch up with waves moving in the same direction, and to leave behind waves travelling in the opposite direction. This causes the wavelength to change.



The Spectra Of Stars

When the ***spectrum*** of the light from a star is analysed, some dark lines are observed. These correspond to colours of light that have been absorbed by substances in the star.

Different substances absorb different colours of light. By identifying the ***wavelengths*** of the colours missing from the spectrum, astronomers can find out which elements are present in the star.

In many cases, missing colours in the spectra of stars are shifted from their expected positions.

- In space, light from the stars appears to be "red shifted" or "blue shifted" due to their movement relative to our galaxy.

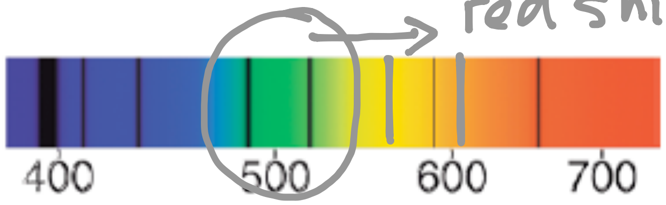
- **Approaching** - ***blue shifted*** (since it has a short wavelength).

  Going away - ***red shifted*** (since it has a long wavelength).

The more the red shift, the faster it moves away.

Increasing energy

red shift



The spectrum of white light from a nearby star. The black lines show which colours have been absorbed by elements in the star. The numbers indicate the wavelength of the light in nanometres.

Increasing wavelength (λ)

- Edwin Hubble found a connection between galactic distance and red shift.
- ***The light from almost all galaxies is red shifted***, and the red shift is greatest for the most distant galaxies.
- This means that the ***speed at which objects are moving away from us is greatest for the most distant objects***.
- Almost everything in the Universe is racing away from us, and the farthest objects are racing away faster than those close to us.
- This seems to be true regardless of the direction in which we make our observations.
- We are not at the centre of the universe - space itself is expanding and every part of it is continuously becoming further away from every other part.
- This observation supports the Big Bang theory for the origin of the universe.

Worksheet 6.4

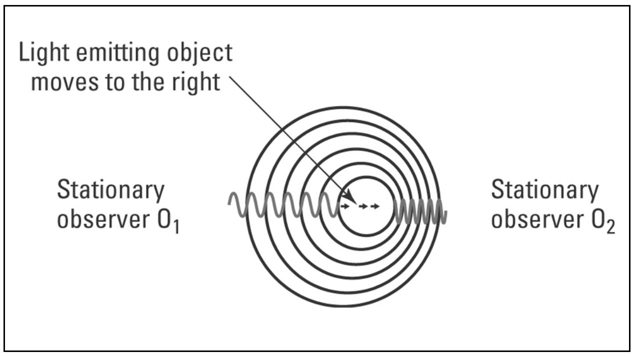
The expanding universe

Student: .... Class: ....

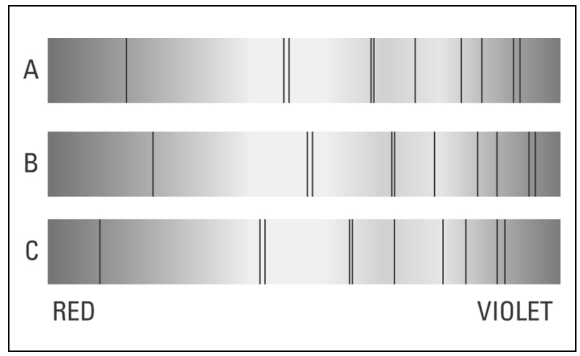
The Doppler effect

1. Figure A shows a light emitting object moving to the right. The light emitted is detected by two stationary observers (O_1 and O_2). Figure B shows absorption spectra of a light emitting object. The dark lines represent light colours that have been absorbed by different elements in the star.

A



B



(a) Explain the Doppler effect using figure A.

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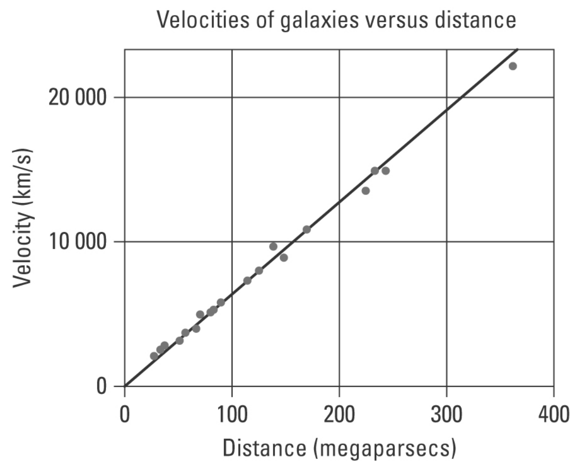
(b) The dark lines in the spectra in figure B show red and blue spectral shifting. One spectrum is of a light source that is stationary relative to the observer. Identify the spectrum (X, Y or Z) in which the light emitting source is moving:

(i) away from the observer ....

(ii) towards the observer. ....

Hubble's expanding universe

2. The following graph shows the velocities of a number of galaxies moving away from our galaxy versus their distance from the Earth.



(a) Describe the trend demonstrated by the graph.

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(b) Explain why these galaxies are moving away from our own.

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THE SUN

The Sun is one of many millions of stars in the universe. It provides energy for Earth in the form of heat and light. This energy can be used for many things.

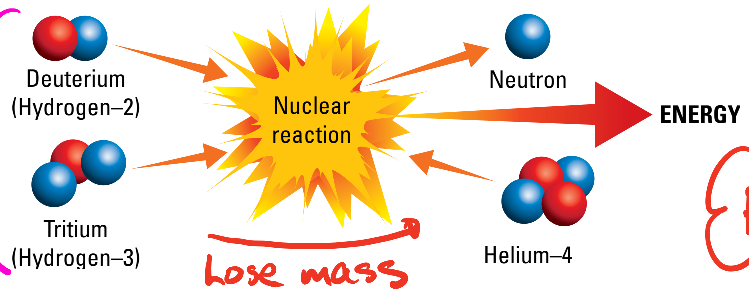
1. Plants use sunlight to make their food to grow by ***photosynthesis***.
2. Animals need the light to see and the heat to keep warm.
3. The heat is responsible for the weather patterns over the Earth's surface.
4. ***Photovoltaic cells*** convert sunlight into electricity.
5. ***Solar hot water heaters*** absorb the heat directly through black collection panels.
6. Houses can be designed to absorb maximum heat in winter and little heat in summer.

Characteristics

Remember this!

1. It is an extremely hot glowing ball of gas (15 million °C in the interior, 6000 °C on the surface).
2. Hydrogen is changed into helium and other elements by a thermonuclear reaction called ***fusion***.

Collide at extreme temperature (~ 1 million °c)



Lose mass

$E = mc^2$

$H \rightarrow He$

3. The Sun is 80% hydrogen, 19% helium, and the remaining 1% is heavier elements.
4. The hot gases exist as ***plasma*** - the fourth state of matter.
5. It is about 150 million kilometres from Earth and is 1 million times bigger in mass.
6. It is about half way through its life and will last for another 5000 million years.

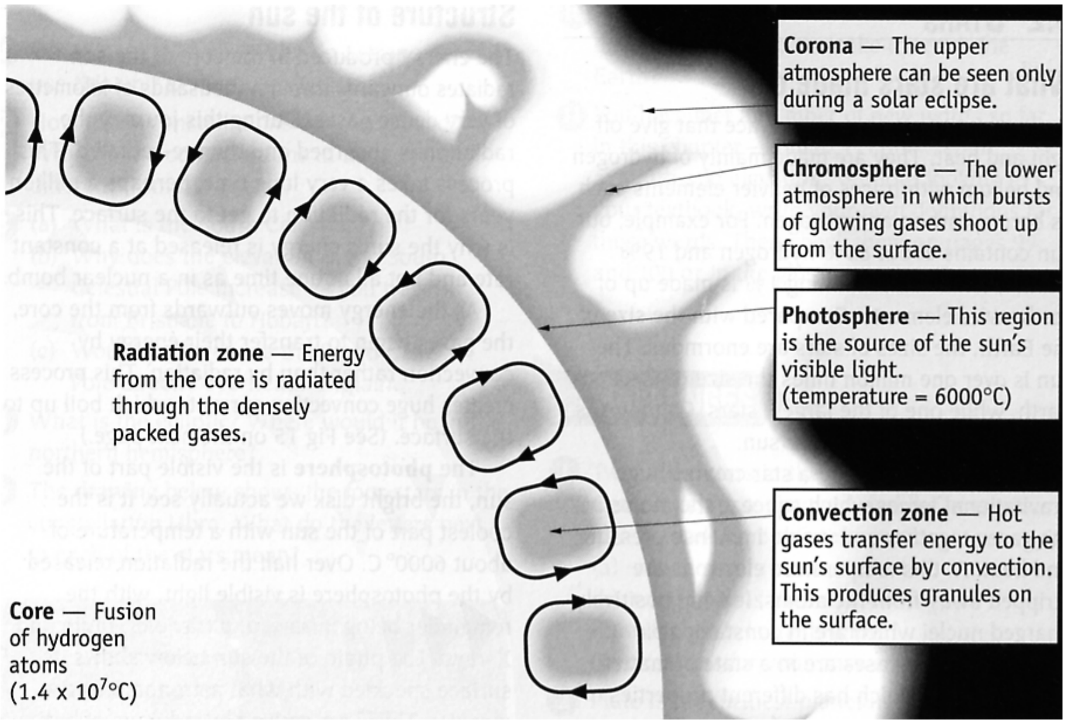
   5 billion years

7. Every second, 5 million tonnes of mass is lost, converted into energy according to Einstein's equation $E = mc^2$.
8. The large mass of the Sun (and other stars) generates a large gravitational force that squeezes the gases together, creating a large amount of heat and pressure. This generates the very high temperatures required for fusion to occur.

Radiations

1. Visible light - enables us to see.
2. Infra-red radiation - can't be seen but is felt as heat.
3. Ultra-violet radiation - causes suntan and skin cancer; helps plants to grow.
4. Radio waves - picked up as "static" on radios.

Structure



Core - thought by many to be made of iron and solid due to the immense pressure.

Photosphere - 300 km thick and made up of dense gas that gives the surface a "grainy" appearance. Is 6000 °C and produces most of the visible light.

Chromosphere - a thin pink-red layer that is only seen during eclipses.

Corona - upper atmosphere of 1 million °C that is clearly seen during an eclipse. Hot gases thrown out from this layer form the "solar wind" of charged particles that sweep outwards through space.

Sunspots

These are black spots that occur on the surface of the Sun at various times. The greatest number appears to occur in an 11 year cycle.

Sunspots are large eruptions that can last for a week or so. This causes an increase in the intensity of the “solar wind”, resulting in a disturbance to radio reception on Earth.

By observing the motion of sunspots across the Sun, astronomers have calculated that the Sun rotates once every 25 days at the equator and one every 34 days near the poles. This indicates that the outer layers are gaseous and not solid.

LIFE CYCLES OF STARS

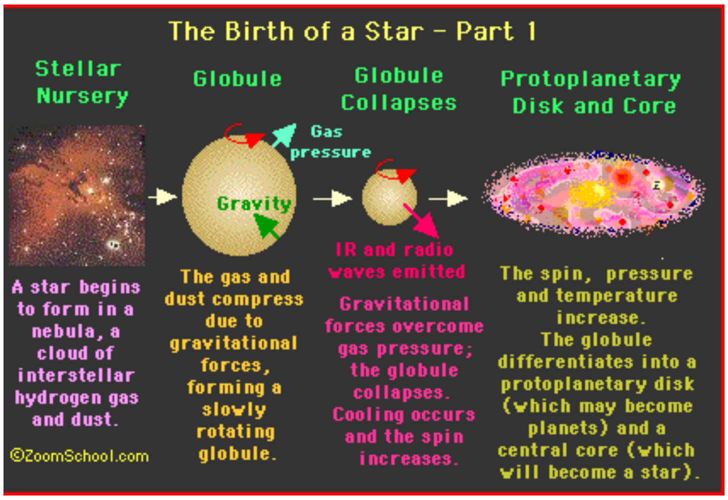
Birth of Stars

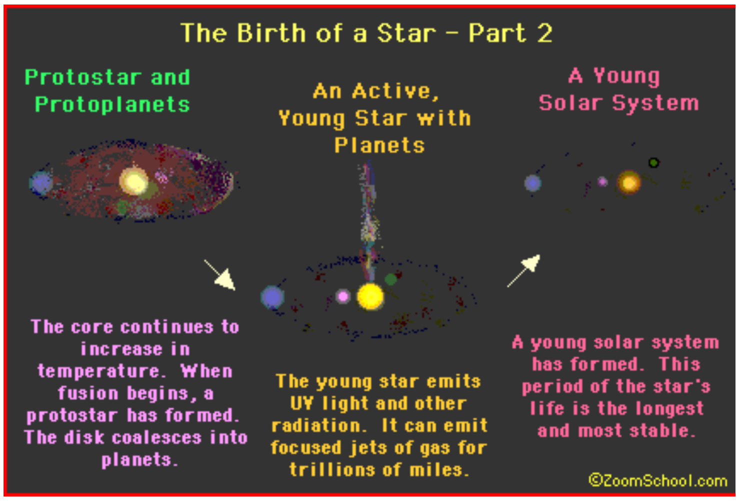
+ heats it

gas cloud + dust

Astronomers believe that stars are born from the gas and dust particles in a ***nebula***. Gravity starts to pull the particles together until the temperature rises, forming a ***protostar***. If the temperature raises high enough, ***nuclear fusion*** starts and the young star begins to glow.

In the fusion process, hydrogen gas is converted into helium gas, releasing huge amounts of energy as matter is annihilated and turned into pure energy according to Einstein's equation ($E = mc^2$).





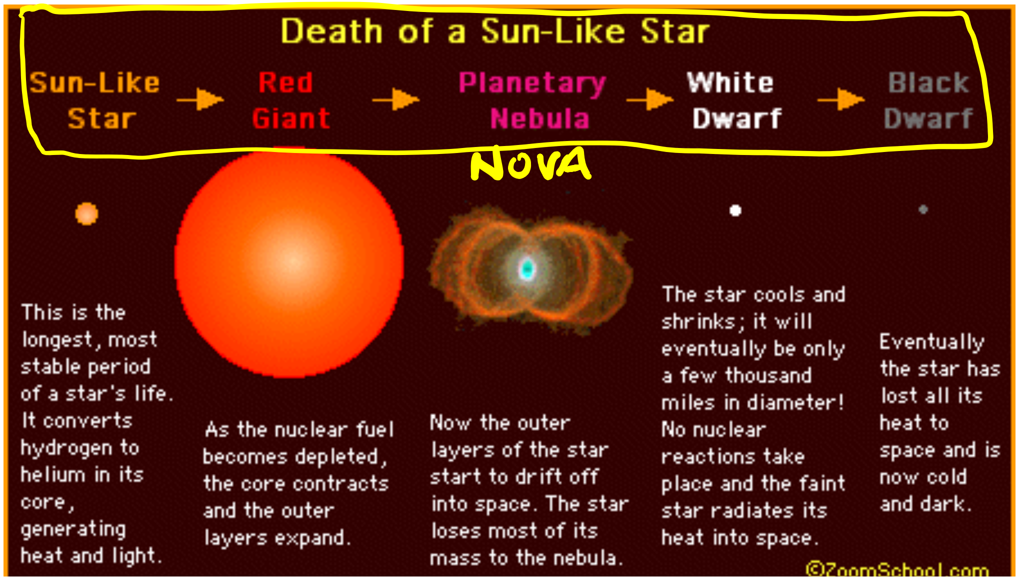
Size

The mass of a star determines its life cycle.

Protostar < 0.1 x mass of our Sun - shrinks but does not get hot enough for fusion to start. Forms a red dwarf before dieing out.

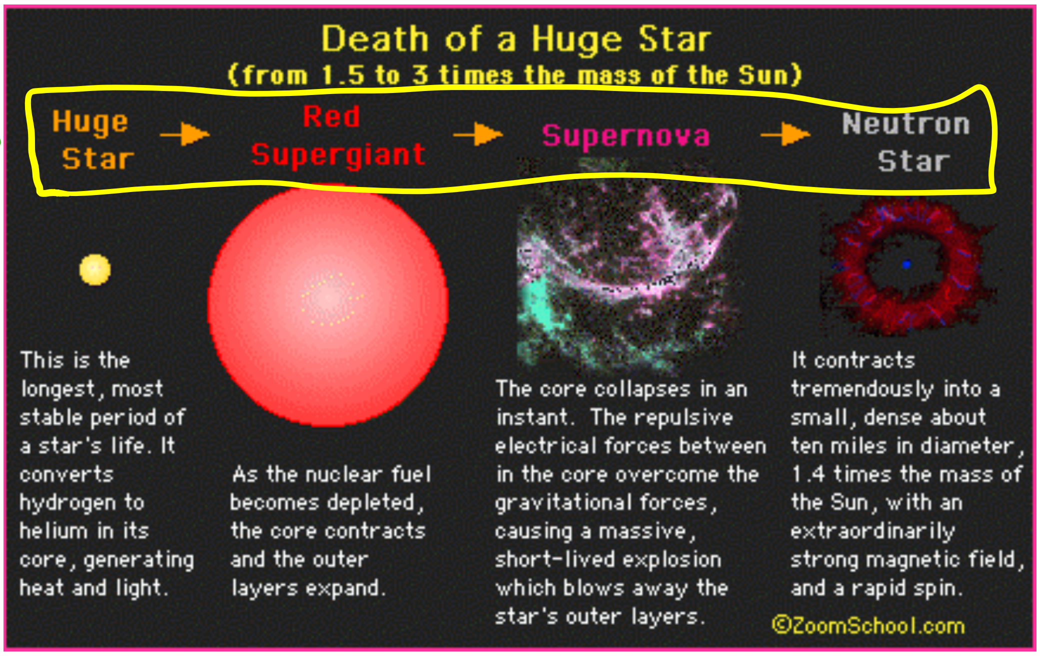
There are three possible lifecycles for stars.

Stars up to 1.5 x mass of Sun - ultimately becomes a ***black dwarf***.

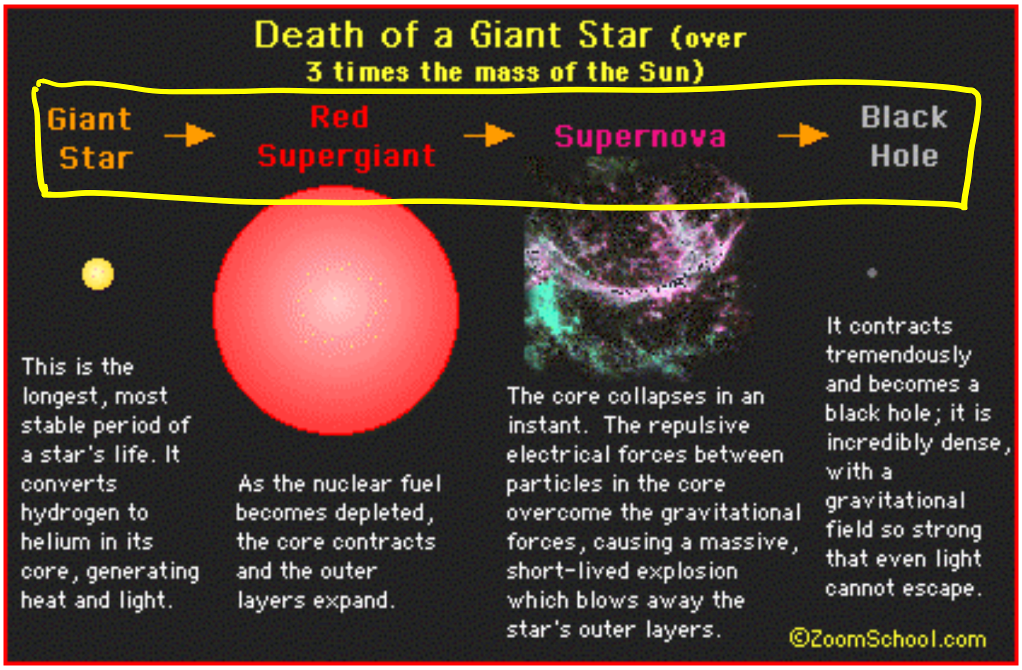


Stars 1.5 - 3.0 x mass of Sun - becomes a ***neutron star***.

Notice the size

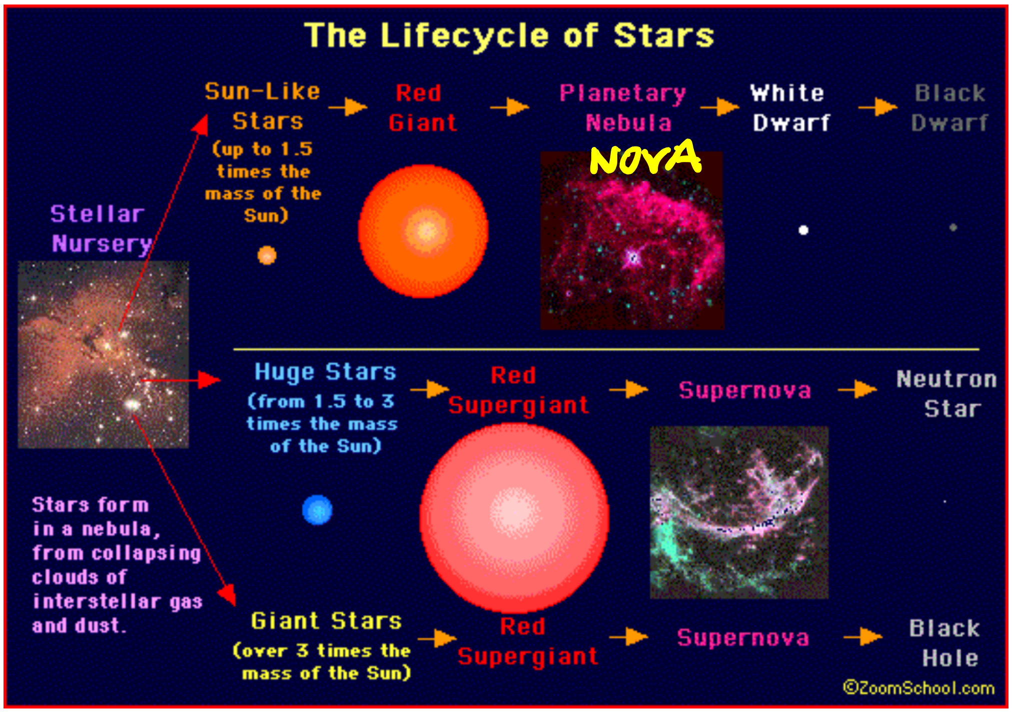


Stars > 3.0 x mass of Sun - becomes a black hole.



Life Cycle Summary

Learn to draw this (see below)

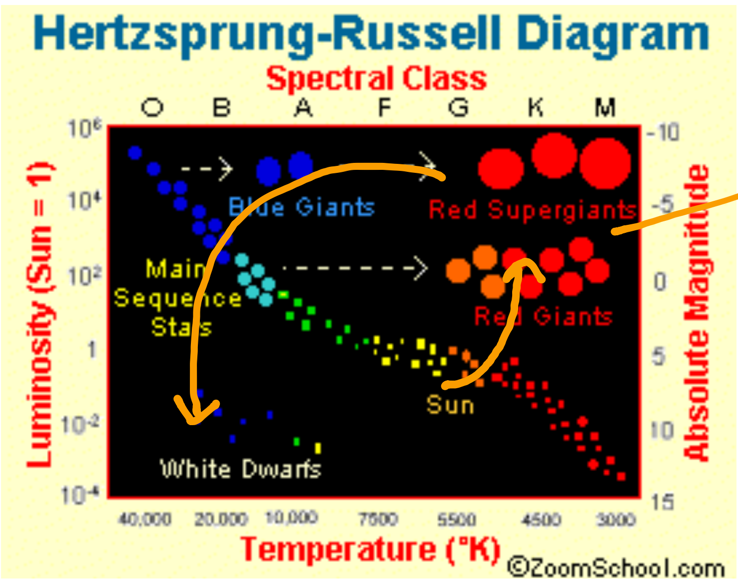


Hertzsprung-Russell Diagram

This diagram plots the absolute magnitude of brightness against the temperature of a star.

Stars that are on the ***main sequence*** are stable and live for billions of years. Our Sun is halfway through its life and has 5 billion years left.

As stars age and die, they evolve and move from the main sequence towards the top of the diagram. After a nova or supernova explosion, stars move down towards the lower left of the diagram.



How stars "evolve" as they die.

Death of stars

Sun ($<1.5x$) → red giant → nova → white dwarf → black dwarf

Huge star ($1.5 \rightarrow 3x$) → red supergiant → supernova → neutron star

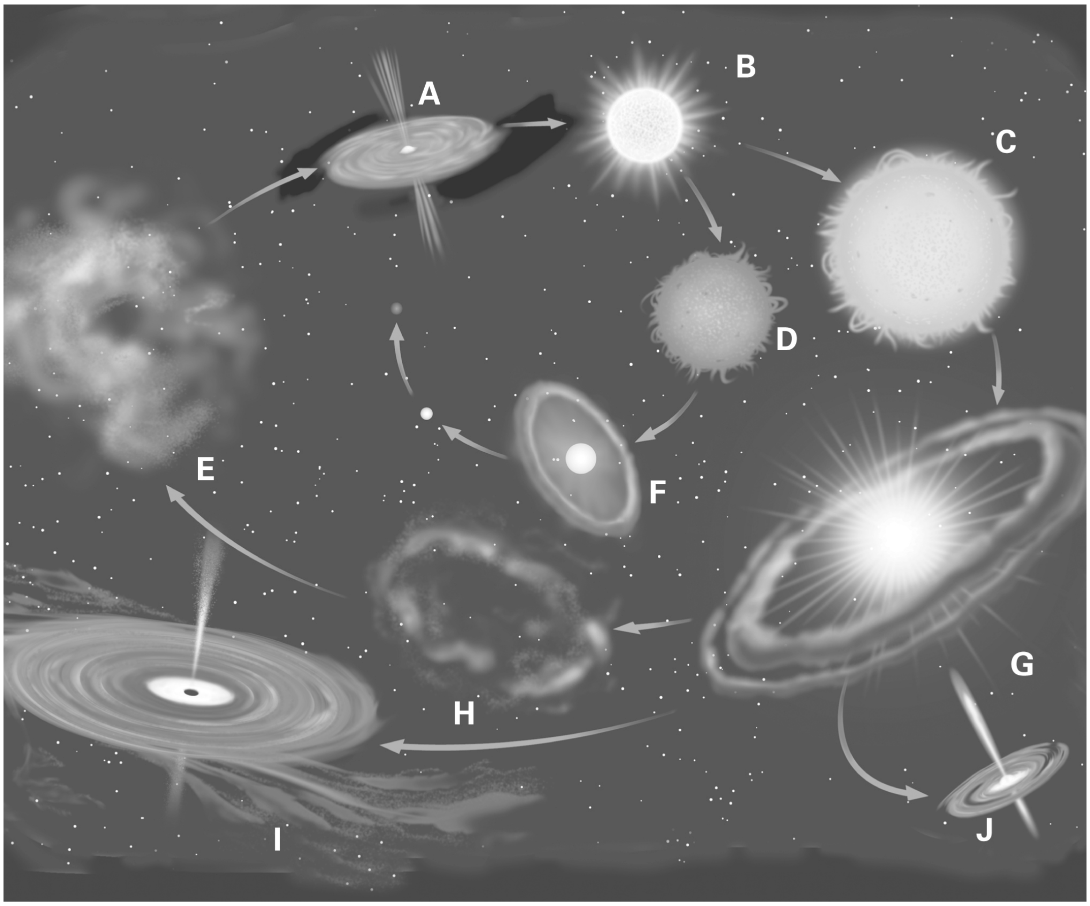
Massive star ($>3x$) → red supergiant → supernova → black hole

Worksheet 6.3

Star life cycles

Student: .... Class: ....

1. The following diagram shows the evolutionary sequence of stars of different masses. Match each of the images, A–J, with the events listed in the table below. These events are randomly listed.



Event	Letter
Star forms and begins to shine.	
Black hole forms from a red super giant explosion (supernova).	
Pulsar (neutron star) forms.	
Red giant star forms from a main sequence star.	
White dwarf forms at the core of a red giant star.	
Supernova formed from a red super giant.	
Protostar forms.	
Red super giant forms from stars heavier than our sun.	
Large nebula forms as gravity pulls dust and gases together.	
Small planetary nebula formed after supernova event.	

2. Read the following text and answer the questions below.

Red Giants

Stars that have a mass between one solar mass and eight solar masses will evolve into red giant stars. During their lives, stars like our sun fuse hydrogen atoms together to form helium. Since helium is denser than hydrogen, the helium that forms sinks to the core of the star. The star 'burns' the hydrogen in the shell that surrounds this core. When the amount of helium in the star reaches about 12%, the core collapses inwards under gravity leading to intense heating. The heat causes the outer layers of the star to expand outwards. The star can swell to one hundred times its original size. The surface of these giant stars begins to cool so that the star now shines with a red light.

(a) Define the term 'one solar mass'.

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(b) Explain why helium is more dense than hydrogen.

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(c) Describe the consequences of the gravitational collapse of the star's core.

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OBSERVING THE NIGHT SKY

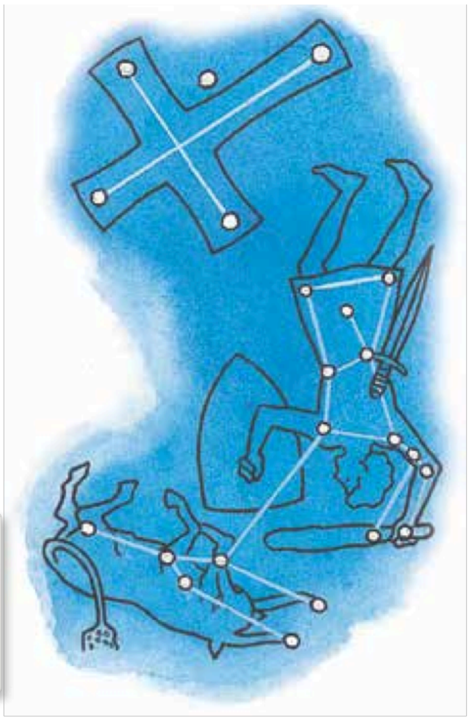
Constellations

Ancient astronomers grouped stars according to the shapes they seemed to form. The shapes were usually of gods, animals or familiar objects.

The most well-known constellations are the 12 groups we know as the signs of the zodiac. These constellations follow the ecliptic (path of the Sun) and their names include Taurus (the bull), Leo (the lion) and Sagittarius (the archer).

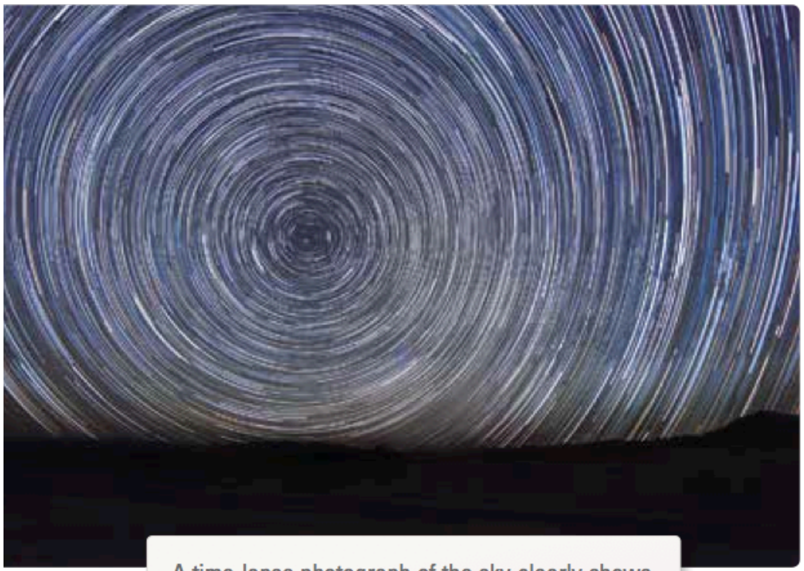
When observed from Earth, the stars in each constellation appear to be very close to each other. But the stars that make up constellations can be located at very different distances from Earth.

e.g. The star Betelgeuse in the Orion constellation is approximately 650 light-years from Earth, whereas the star Bellatrix in the same constellation is about 240 light-years from us.



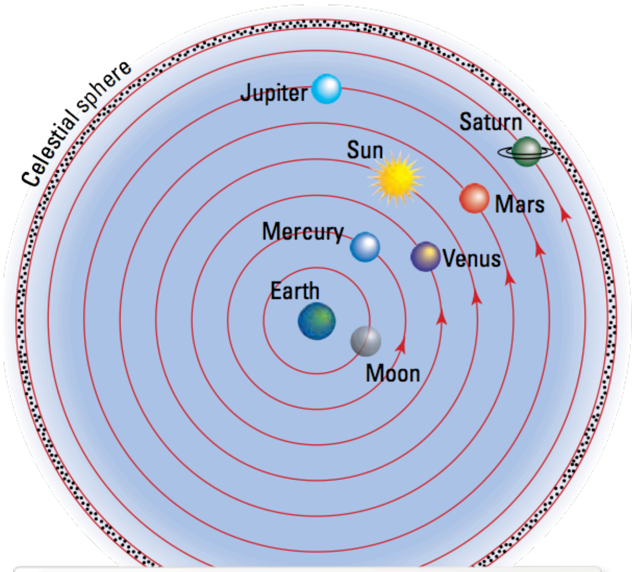
The constellations Crux, Orion and Taurus as seen from Australia

The apparent motion of the fixed pattern of stars at night, shown in the time-lapse photograph below, is due to the rotation of the Earth.



A time-lapse photograph of the sky clearly shows the apparent movement of the stars.

The apparent change in position of the constellations is due to the Earth's orbit around the sun. Sky charts, sometimes called sky maps, star maps or star charts, show the position of constellations, stars and the planets from different locations for each month of the year.



In 150 AD, the Greek astronomer Ptolemy suggested that the stars were attached to a 'celestial sphere' that rotated above our heads. According to Ptolemy, the sun, the planets and the Moon also orbited the Earth.

Observations using telescopes showed that many different types of objects in the sky could be identified. These included ***single or double stars, groups of stars called galaxies, clusters of galaxies, and clouds of gas and dust called nebulae***.

Galaxies

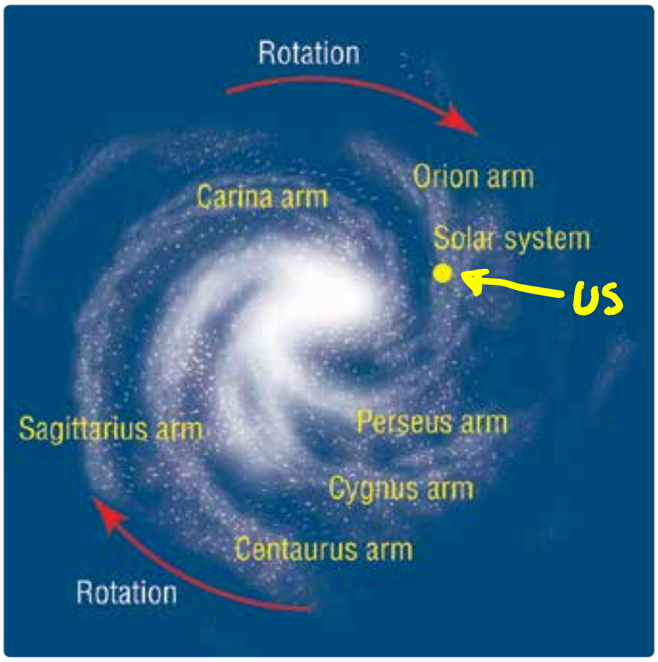
Stars group together in groups to form galaxies, attracted towards each other by gravitational forces.

Our own sun is one of an estimated 200 - 400 billion stars in the Milky Way galaxy.

We think there are more than 100 billion other galaxies of different sizes and shapes throughout the universe.

Each of these galaxies is home to stars at all stages of their life cycles.

The Milky Way galaxy, shown at right, is a spiral galaxy.



Our solar system is found on the Orion arm of the spiral. Due to the rotation of the galaxy, our solar system orbits the centre of the galaxy at a speed of about 200 kilometres per second.

Worksheet 6.1

Observing stars

Student: .... Class: ....

1. Three types of celestial objects are illustrated below. Identify and outline the nature of each.

(a)

Name: ....
Outline: ....
....
....
....
....
....
....
....
....

(b)

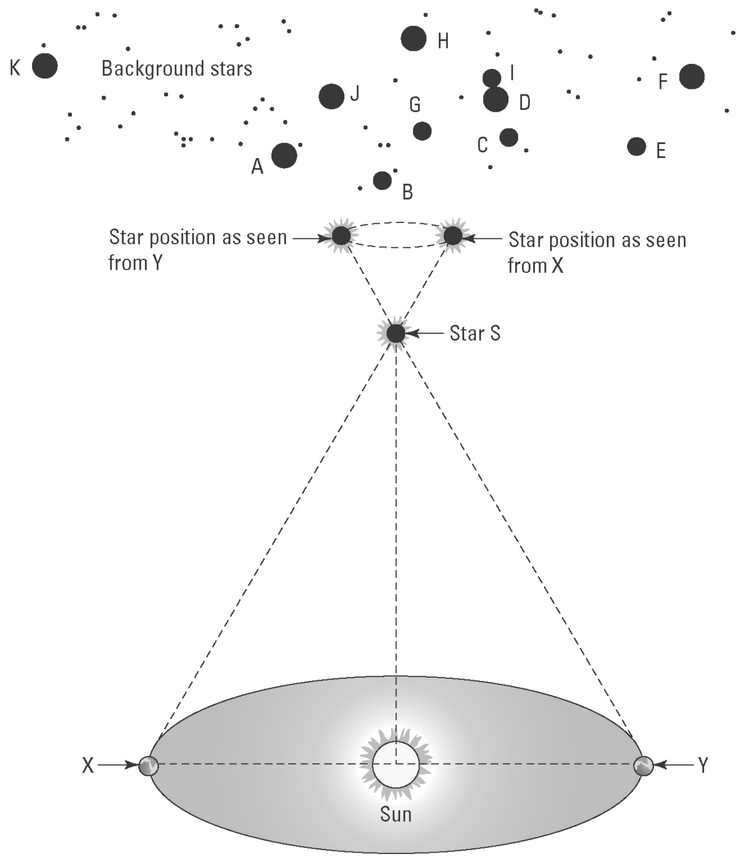
Name: ....
Outline: ....
....
....
....
....
....
....
....

(c)

Name: ....
Outline: ....
....
....
....
....
....
....
....

Photos:
Top: creativemarc / 123RF
Centre: NASA, Hubble Heritage Team, STScI
Bottom: Yuri Beletsky / NASA

2. The following diagram shows the effect of parallax on the observation of a close star (S) from two locations (X and Y) in the Earth's orbit around the sun.



(a) The Earth is shown in two positions X and Y. How long does it take for the Earth to move from X to Y?

....

(b) Explain the meaning of parallax in relation to the apparent motion of stars.

....

....

(c) Before 1837 it was believed that all stars existed on a transparent celestial sphere at a fixed distance from the Earth. How did the observation of parallax of close stars change this view?

....

....

....

....

(d) Outline the motion of the stars as seen by an observer on Earth during the time taken to travel from X to Y.

....

....

....

....

....

LIGHT FROM STARS

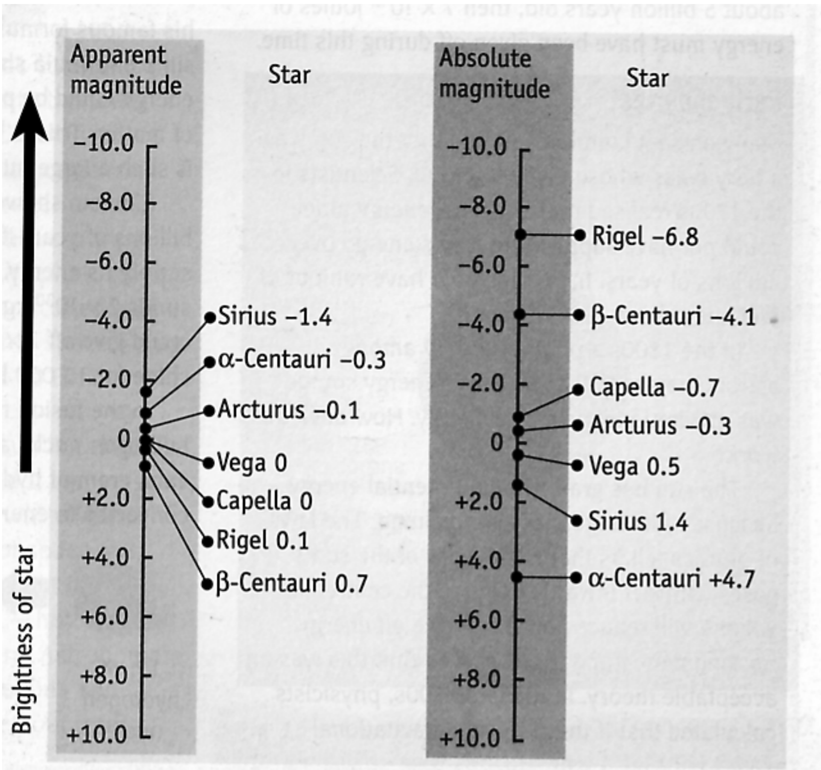
Brightness

- Measured on a scale called ***apparent magnitude*** when viewed from Earth.
- It does not measure the actual brightness of a star as the distance from the Earth will change how bright a star appears.

  i.e. A very bright star far away will appear very dim from Earth.

- The actual brightness is measured by the ***absolute magnitude*** scale - this scale measures the amount of light given off by a star if it is a set distance from Earth.
- For both scales, ***a negative number means the star is very bright***.
- The amount of light given off by a star is determined by its size or amount of matter in it.

  e.g. Canopus is 80,000 times more massive than our Sun so it gives off much more light - it has a greater and more negative absolute magnitude.



- The absolute magnitude scale is the one used by astronomers to compare stars.

red - cool
yellow - average
blue - hot

white = hot

Remember!

Colour

- Colours of stars vary from ***red (cool) to blue (hot)***.
- Astronomers use Wien's Law to calculate the temperature of stars by measuring the wavelength of light given off.

i.e. Wien's Law:

$$\boxed{\text{temperature} = \frac{3 \times 10^6}{\text{wavelength}}}$$

where wavelength is measured in nanometres (x 10^{-9} m).

Examples

Two stars with the same colour have the same temperature. If one star has a greater absolute magnitude (brightness), it must be bigger.

A red and blue star can have the same size and absolute magnitude (brightness), but the blue star radiates more energy because it is hotter.

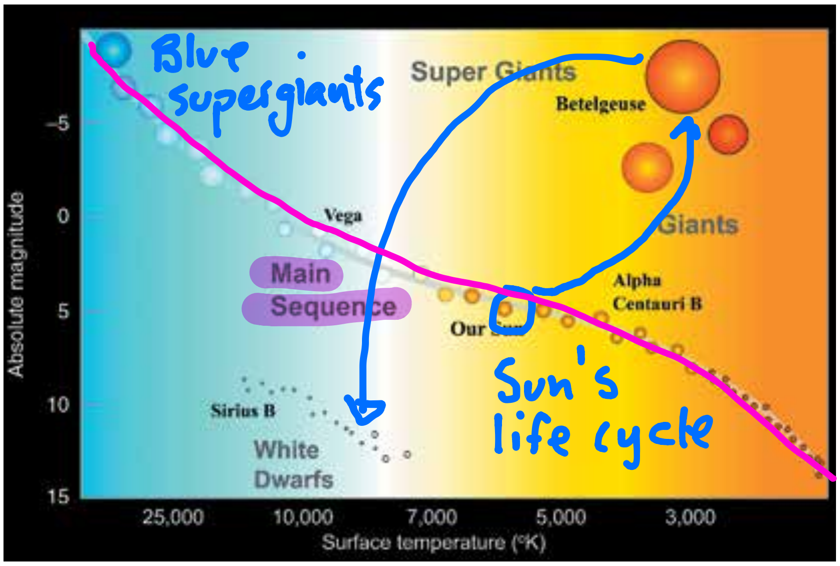
Colour and brightness

An interesting way of displaying the data collected about stars was developed independently by two astronomers, Ejnar Hertzsprung from Denmark and Henry Norris Russell from America. This method has now been named after both of them.

In the Hertzsprung–Russell diagram, the absolute brightness of a star is plotted against its surface temperature, which is deduced from its colour.

When data for many stars are plotted, most of them, including our sun, fall into what is known as ***the main sequence***.

Exactly where a star is found along the main sequence is determined by its mass.



Low-mass stars tend to be cooler and less bright than high-mass stars.

Worksheet 6.6

The mysterious universe: Summary

Student: .... Class: ....

Use the words listed below to complete the following sentences.

absolute	protostar	electrons	spiral	observer
constellations	space	giants	billion	radio
fusion	time	nuclei	expansion	temperature
nebulae	afterglow	pulsars	helium	

1. Groups of stars form patterns that are called ....

2. Our solar system is part of a large .... galaxy called the Milky Way.

3. Clouds of dust and gas in interstellar space are called ....

4. The apparent movement of stars at different distances is caused by the movement of the .... This effect is called parallax.

5. According to the big bang theory, the universe is about 13.7 .... years old.

6. The first stage of the big bang involved the formation of .... and space. The early universe was very small and very hot. Space then underwent rapid .... and it cooled as it expanded.

7. Quarks and .... were the first matter particles to form. Subsequent cooling led to proton and neutron formation, which then combined to form ....

8. Red shifting of lines in star spectra is consistent with the stretching of .... as the universe expands.

9. The big bang theory can explain the large amount of .... relative to hydrogen in the universe.

10. Microwave background radiation has been detected by instruments aboard satellites. This microwave radiation is the cooled .... of the big bang.

11. Stars are formed when gravity causes gas and dust to come together and heat up. Eventually a .... is formed and then finally a star of a given mass.

12. Stars like our sun evolve into red .... before they explode.

13. Red supergiants evolve into black holes or ....

14. The .... magnitude of a star is a measure of its real brightness when all stars are compared at the same distance.

15. The Hertzsprung–Russell diagram plots the surface .... of the star versus its absolute magnitude. This diagram helps us understand the evolutionary life cycle of a star.

16. Stars generate their energy by nuclear .... in which hydrogen nuclei join together to form heavier nuclei.

17. The universe can be studied using optical (visible light) telescopes as well as .... telescopes.

Worksheet 6.1

Observing stars

Answers

1. Three types of celestial objects are illustrated below. Identify and outline the nature of each.

(a)

Name: Nebula

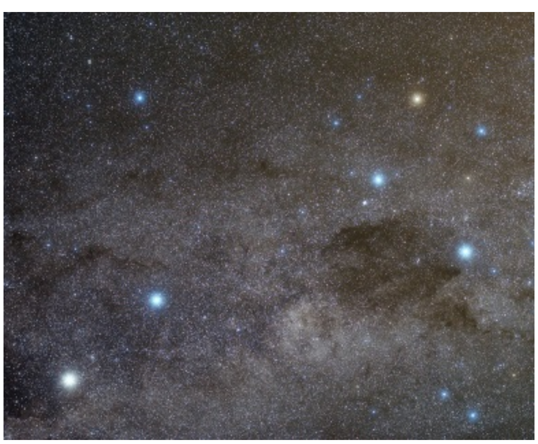
Outline: Clouds of gas and dust that can come together and heat up to form stars

(b)

Name: Galaxy

Outline: A group of stars and gas held together by gravity

(c)

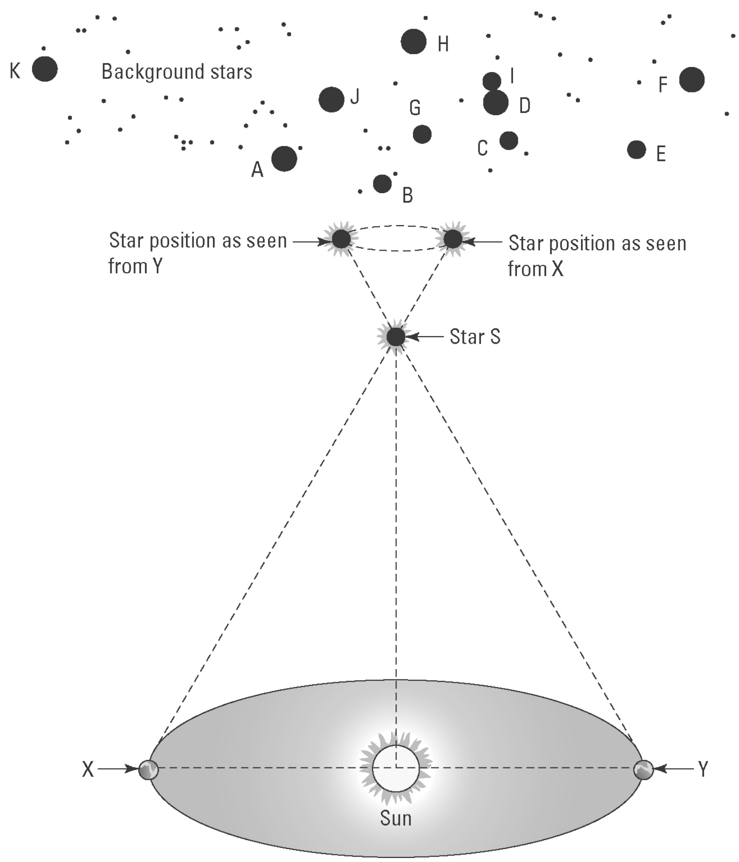


Name: Constellation

Outline: A region of stars that form a pattern when viewed from Earth

Photos:
Top: creativemarc / 123RF
Centre: NASA, Hubble Heritage Team, STScI
Bottom: Yuri Beletsky / NASA

2. The following diagram shows the effect of parallax on the observation of a close star (S) from two locations (X and Y) in the Earth's orbit around the sun.



(a) The Earth is shown in two positions X and Y. How long does it take for the Earth to move from X to Y?

6 months..

(b) Explain the meaning of parallax in relation to the apparent motion of stars.

The apparent movement of objects at different distances due to the motion of the observer

....

(c) Before 1837 it was believed that all stars existed on a transparent celestial sphere at a fixed distance from the Earth. How did the observation of parallax of close stars change this view?

The stars must be at different distances from the Earth otherwise parallax effects would not be possible.

....

(d) Outline the motion of the stars as seen by an observer on Earth during the time taken to travel from X to Y.

Star S appears to move relative to the fixed background stars. S will appear to the right of B, G and H from point X but 6 months later when viewed from point Y, star S will appear from the left of B, G and H.

....

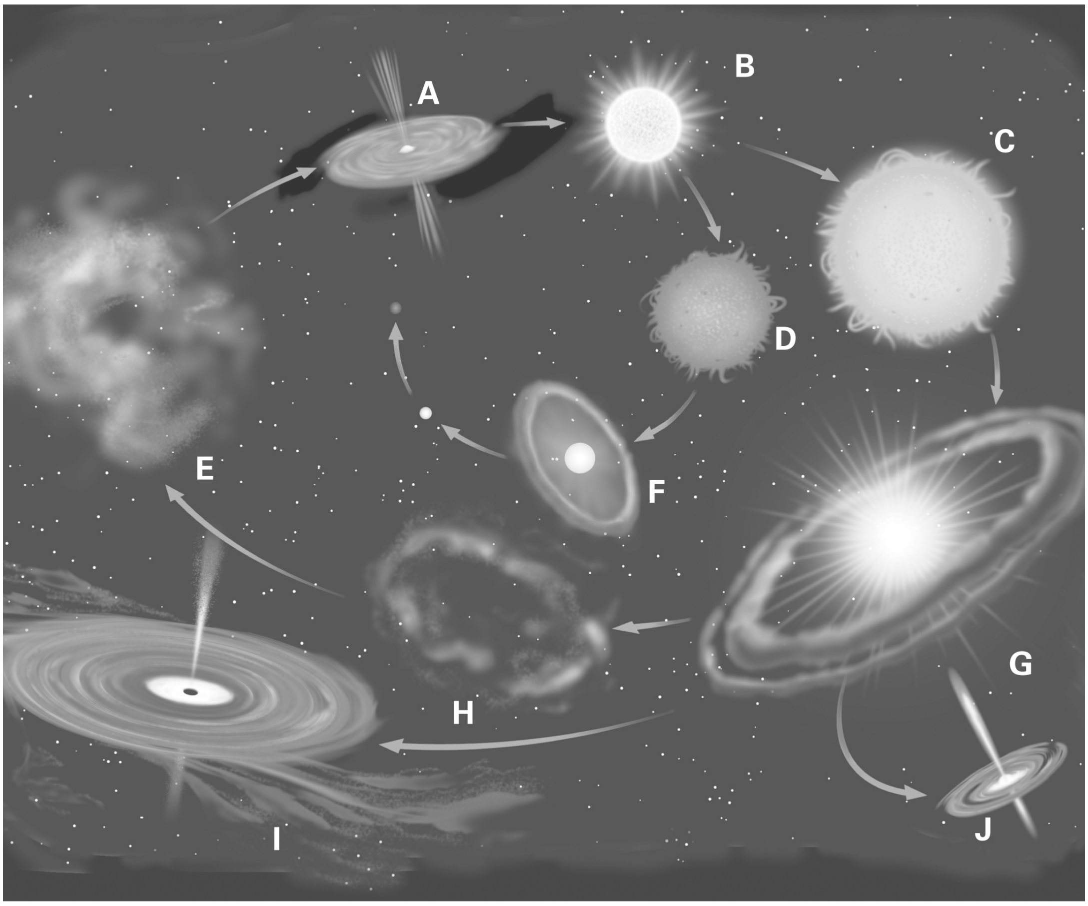
....

....

Star life cycles

Answers

1. The following diagram shows the evolutionary sequence of stars of different masses. Match each of the images, A–J, with the events listed in the table below. These events are randomly listed.



Event	Letter
Star forms and begins to shine.	B
Black hole forms from a red super giant explosion (supernova).	I
Pulsar (neutron star) forms.	J
Red giant star forms from a main sequence star.	D
White dwarf forms at the core of a red giant star.	F
Supernova forms from a red super giant.	G
Protostar forms.	A
Red super giant forms from stars heavier than our sun.	C
Large nebula forms as gravity pulls dust and gases together.	E
Small planetary nebula formed after supernova event.	H

2. Read the following text and answer the questions below.

Red giants

Stars that have a mass between one solar mass and eight solar masses will evolve into red giant stars. Betelgeuse in the Orion constellation is a red giant star. During their lives, stars like our sun, fuse hydrogen atoms together to form helium. Since helium is denser than hydrogen, the helium that forms sinks to the core of the star. The star 'burns' the hydrogen in the shell that surrounds this core. When the amount of helium in the star reaches about 12%, the core collapses inwards under gravity leading to intense heating. The heat causes the outer layers of the star to expand outwards. The star can swell to one hundred times its original size. The surface of these giant stars begins to cool so that the star now shines with a red light.

(a) Define the term 'one solar mass'.

One solar mass is the mass of our sun.

(b) Explain why helium is more dense than hydrogen.

Helium has a heavier nucleus (2 protons and 2 neutrons) than hydrogen (1 proton).

(c) Describe the consequences of the gravitational collapse of the star's core.

The core of helium collapses due to the inward pull of gravity, resulting in rapid heating. This intense heat then causes the expansion of the hydrogen shell outwards, producing a large red giant star.

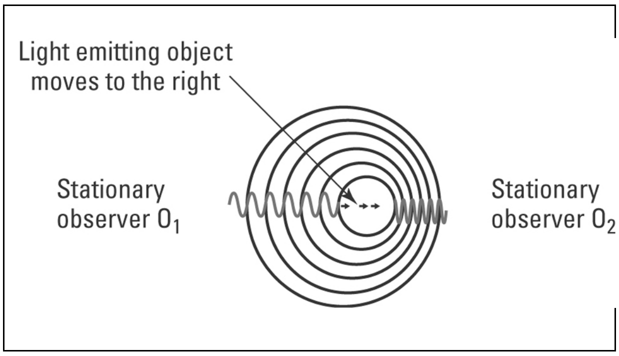
The expanding universe

Answers

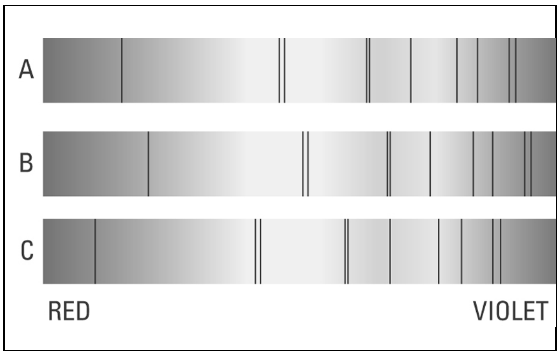
The Doppler effect

1. Figure A shows a light emitting object moving to the right. The light emitted is detected by two stationary observers (O_1 and O_2). Figure B shows absorption spectra of a light emitting object. The dark lines represent colours of light that have been absorbed by different elements in the star.

A



B



(a) Explain the Doppler effect using figure A.

Observer 1 sees the light source moving away from them. The light waves emitted from this source spreads out, increasing the wavelength. This is seen as a red shift in the light. Observer 2 sees the light source coming towards them and the light waves bunch up, decreasing the wavelength. This is seen as a blue shift in the light.

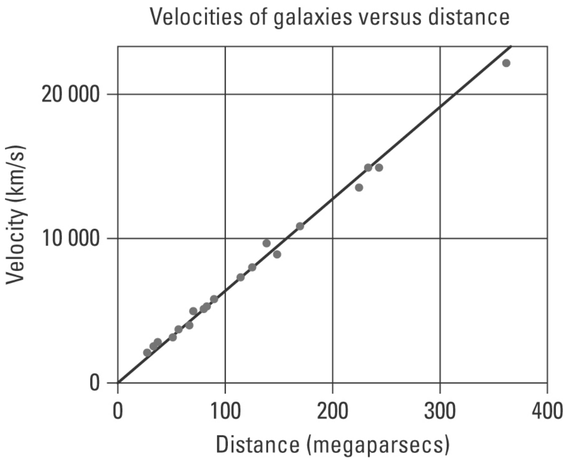
(b) The dark lines in the spectra in figure B show red and blue spectral shifting. One spectrum is of a light source that is stationary relative to the observer. Identify the spectrum (X, Y or Z) in which the light emitting source is moving:

(i) away from the observer C

(ii) towards the observer. B

Hubble's expanding universe

2. The following graph shows a plot of the velocities of a number of galaxies moving away from our galaxy versus their distance from the Earth.



(a) Describe the trend demonstrated by the graph.

The greater the distance a galaxy is from the Earth, the greater its velocity as it moves away from Earth, i.e. the further away the galaxy the faster it is moving.

(b) Explain why these galaxies are moving away from our own.

The universe has been expanding since the big bang and continues to expand today. Therefore, these galaxies, including our own, are moving away from each other.

Worksheet 6.5

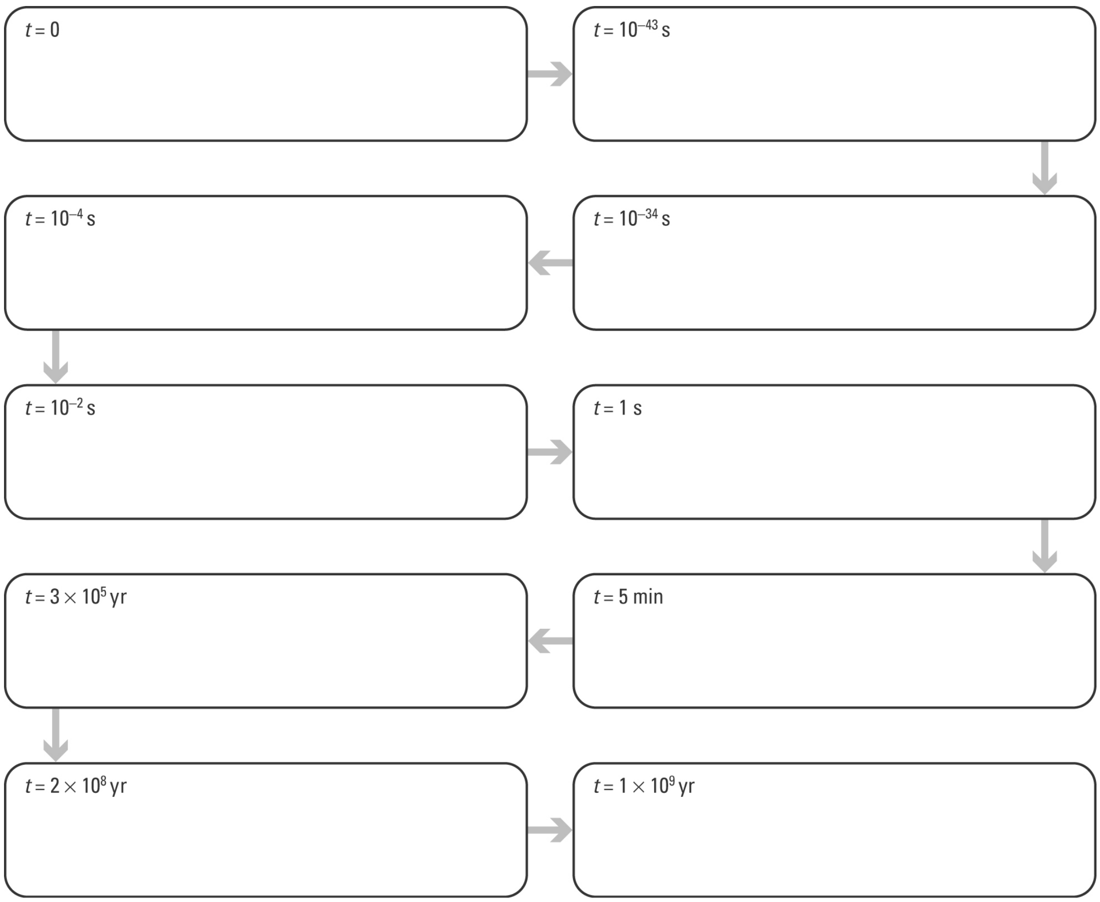
The big bang

Answers

Flow charting the big bang

1. Complete the big bang flow chart by selecting the event from the list below that matches each step in the sequence and rewriting it in the correct box. Enter a title for each box that summarises each event.

Correct order = E, C, I, D, F, J, G, A, H, B



Big bang events in random order:

A. Universe has cooled to 3000 °C; first atoms form.

B. Galaxies begin to form.

C. Time and space form; space expands rapidly.

D. Heavier matter forms — protons, neutrons.

E. Singularity exists — the big bang occurs.

F. Universe cools; reaches size of the solar system.

G. Atomic nuclei form.

H. First stars appear.

I. Light matter forms — electrons, positrons.

J. Universe has cooled to 10^{10} °C.

Evidence for the big bang

2. The following microwave map of the early universe (t = 380 000 years after big bang) was produced by a probe that orbited the Earth in 2001.

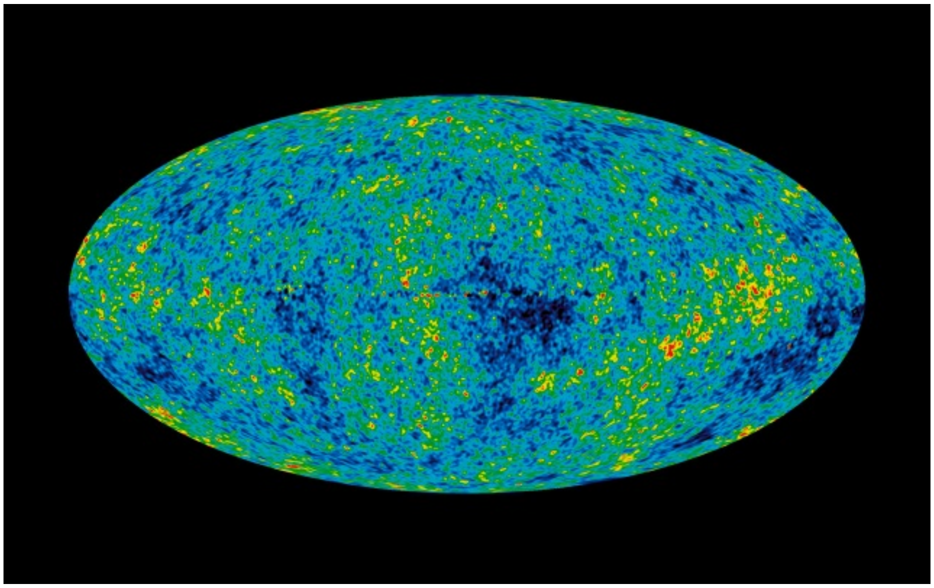


Photo: NASA

(a) The map shows that in the early universe matter was not uniformly distributed. Explain how this map supports the big bang theory.

The big bang theory states that the very early universe was very hot and that as it cooled matter particles started to form. As further cooling occurred, gravity caused the matter to condense and form lumps, eventually forming galaxies. The cosmic microwave background radiation map shows that the matter is not evenly spread, which is consistent with the expanding universe model.

(b) In 1948, Gamow and Alpher predicted that microwave radiation should now fill the universe. They proposed that this radiation was the cooled 'afterglow' of the big bang. When was this microwave radiation first detected?

1965

The mysterious universe: Summary

Answers

Use the listed words to complete the sentences.

absolute	protostar	electrons	spiral	observer
constellations	space	giants	billion	radio
fusion	time	nuclei	expansion	temperature
nebulae	afterglow	pulsars	helium	

1. Groups of stars form patterns that are called ....constellations.........
2. Our solar system is part of a large ....spiral............. galaxy called the Milky Way.
3. Clouds of dust and gas in interstellar space are called ....nebulae................
4. The apparent movement of stars at different distances is caused by the movement of the ....observer.... This effect is called parallax.
5. According to the big bang theory, the universe is about 13.7 ....billion.......... years old.
6. The first stage of the big bang involved the formation of ....time........... and space. The early universe was very small and very hot. Space then underwent rapid ....expansion..... and as it expanded it cooled.
7. Quarks and ...electrons... were the first matter particles to form. Subsequent cooling led to proton and neutron formation, which then combined to form .....nuclei..............
8. Red shifting of lines in star spectra is consistent with the stretching of ....space......... as the universe expands.
9. The big bang theory can explain the large amount of ....helium....... relative to hydrogen in the universe.
10. Microwave background radiation has been detected by instruments aboard satellites. This microwave radiation is the cooled ....afterglow..... of the big bang.
11. Stars are formed when gravity causes gas and dust to come together and heat up. Eventually a ....protostar........ is formed and then finally a star of a given mass.
12. Stars like our sun evolve into red ......giants......... before they explode.
13. Red supergiants evolve into black holes or ....pulsars........
14. The ....absolute................ magnitude of a star is a measure of its real brightness when all stars are compared at the same distance.
15. The Hertzsprung–Russell diagram plots the surface ...temperature.... of the star versus its absolute magnitude. This diagram helps us understand the evolutionary life cycle of a star.
16. Stars generate their energy by nuclear ......fusion.......... in which hydrogen nuclei join together to form heavier nuclei.
17. The universe can be studied using optical (visible light) telescopes as well as ......radio.......... telescopes.